Second Response to the Referee Report on "Comments on Celestial CFT and AdS_3 String Theory"

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I. PREAMBLE

I thank the Editor for sending my manuscript for a second review, and I also owe my thanks to the Referee for studying my manuscript carefully and for recommending the work for publication. Seeing my work studied meticulously is one of my principal sources of motivation, and it was therefore a pleasure to prepare this response to the referee's second report.

I am especially grateful to the Referee for encouraging me to include in my manuscript the discussion of the WZNW model that I submitted in response to the first round of peer review, which, although familiar to experts in WZNW models, may not yet have been carefully presented in the literature concerning dynamical models for perturbative celestial CFT. I also decided to address, from a new perspective, a point that the referee raised in the first report, which I discussed in the final section of my first response and which concerns the Weyl anomaly.

After submitting my first response, I reflected more carefully on the question of the Weyl anomaly. I then realised that there is an argument due to K. Nguyen that fits very well with my proposal. This argument elucidates very clearly the differences in physical interpretation between the central charge of the boundary-spacetime Virasoro algebra, originating from the string-theory construction in Euclidean AdS_3, and the physical celestial stress-energy tensor, appertaining to the celestial CFT. I therefore decided to add a new discussion of this issue.

I also recognise that some spelling and typographical errors in the original version of the manuscript needed to be corrected, and I have corrected them in the new version utilising a local installation of a program called "LanguageTool," which is suitable for texts written in LaTeX.

Finally, in the new version, I have also included citations to the following works:

> Donnay, Laura, et al. "Celestial open strings at one-loop." **Journal of High Energy Physics** 2023.10 (2023): 1–20.
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> Jiang, Hongliang. "Celestial OPEs and $w_{1+\infty}$ algebra from worldsheet in string theory." **Journal of High Energy Physics** 2022, no. 1 (2022): 101.
>
> Kervyn, Xavier, and Stephan Stieberger. "High energy string theory and the celestial sphere." **Journal of High Energy Physics** 2025, no. 9 (2025): 1-27.
>
> Stieberger, S., & Taylor, T. R. (2018). Strings on celestial sphere. **Nuclear Physics B**, 935, 388-411.

Although these works differ from mine in the particular way in which string theory is employed to study celestial CFT, I believe that these works and the manuscript under review belong to the broader and, at the same time, ambitious research programme of utilising string theory to extract information about the putative celestial CFT.

Once again, I thank the referee for reading my work carefully, and I thank the Editor for inviting me to respond to the second referee report before formalising the manuscript's acceptance for publication.

II. MORE COMMENTS ON THE WEYL ANOMALY

On submitting the first reply to the referee report, I considered more carefully the puzzle concerning the difference between the central charge of the boundary-spacetime Virasoro algebra, originating from the string-theory construction on Euclidean AdS_3, and the fact that, under a set of boundary conditions and flux prescriptions, the classical BMS flux algebra can be shown to close under the standard Peierls bracket without a central term (see Donnay and Ruzziconi [1]).

I subsequently realised that there is an important observation, due to Nguyen [2], that the geometry underlying the holographic uplifting of the AdS_3/CFT_2 correspondence to flat-space holography, in the sense of Ball *et al.* [3], does not compel us to identify the boundary-spacetime Virasoro stress-energy tensor of the string-theory construction with the physical celestial stress-energy tensor. Therefore, according to this argument, no contradiction follows merely from the difference between their central charges.

Given the importance of this observation to the proposal advanced in the manuscript, namely, the construction of a candidate model for the perturbative celestial CFT approximating the tree-level MHV sub-sector of Einstein gravity around flat space, I decided to add the following discussion as subsection 1.3, "The Puzzle of the Central Charge," of the revised version of the manuscript:

> Now we should address the following puzzle. First, recall that the existence of the extended BMS algebra (see Barnich and Troessaert [4]), possessing two centre-less Virasoro sub-algebras, served as a motivation (see Pasterski and Shao [5]) for the reformulation of the scattering amplitudes in perturbative quantum gravity as the correlation functions of a putative CFT residing on the celestial sphere at null infinity, referred to as the "celestial CFT." A significant consequence of this reformulation was the re-interpretation (see Kapec *et al.* [6] and Kapec *et al.* [7]) of the sub-leading soft gravity theorem as a conformal Ward identity for the physical celestial stress-tensor $\boldsymbol{T}^{\mathrm{CCFT}}_{zz}$.
>
> On the other hand, in the present manuscript, we shall propose that the large-level limit of the correlation functions of a set of appropriately dressed vertex operators, constructed from the conformal primary fields of the H_3^+-WZNW model, reproduces the tree-level MHV leaf amplitudes for gravitons, thereby suggesting a candidate model for the perturbative celestial CFT approximating the tree-level MHV sub-sector of general relativity defined perturbatively around flat space. However, the boundary-spacetime Virasoro algebra realised by the string theory construction on Euclidean AdS_3, which resides in the boundary of the conformal completion of the three-dimensional Lobachevsky space H_3^+, has a non-zero central charge related to the three-dimensional Newton constant.
>
> Therefore, we must clarify how our proposal is consistent with the fact that, as argued by Donnay and Ruzziconi [1], under a set of boundary conditions and flux prescriptions, the classical BMS flux algebra can be shown to close under the standard Peierls bracket without a central term, and that the corresponding celestial stress-energy tensor OPE has a vanishing Virasoro central charge.
>
> Our answer to this question is based on an observation due to Nguyen [2], and in order to explain it precisely, we should briefly recall some geometric notions about

asymptotically flat spacetimes.

Let u denote a coordinate function associated to a generator of null infinity $\mathscr{I}$, and consider an oriented system of real coordinates x^A charting the corresponding celestial cuts. We may accordingly represent the metric tensor on the celestial cuts by $q_{AB} = \mathrm{e}^{2\psi} q_{0,AB}$, where $q_{0,AB}$ is a reference metric possessing constant curvature, whereas $\partial_u \psi = 0$. In addition, let D_A be the Levi-Civita covariant derivative operator, metrically compatible with q_{AB}, and for any given tensor X_{AB}, we define its trace-free part by

$$(X_{AB})^{\mathrm{TF}} \coloneqq X_{AB} - \frac{1}{2} q_{AB} q^{CD} X_{CD} \,. \tag{1}$$

We now introduce the lift to null infinity of a symmetric, trace-free tensor residing on the celestial cuts, given by

$$\mathcal{T}_{AB} \coloneqq 2\,(\mathrm{D}_A \mathrm{D}_B \psi + \mathrm{D}_A \psi\, \mathrm{D}_B \psi)^{\mathrm{TF}} \,, \quad \mathcal{T}_{u\mu} \coloneqq 0 \,.$$

Thus, given a local complex conformal coordinate z charting a coordinate neighbourhood of the celestial sphere, in terms of which $q_{AB}\mathrm{d}x^A \mathrm{d}x^B = \mathrm{e}^{2\sigma} \mathrm{d}z\mathrm{d}\bar{z}$, we have

$$\mathcal{T}_{zz} = 2\,(\partial_z^2 \psi - 2(\partial_z \sigma)(\partial_z \psi) + (\partial_z \psi)^2) \,. \tag{2}$$

Next, let $\boldsymbol{N}^{zz}$ denote the covariant news tensor with indices raised with respect to q_{AB}, so that $\boldsymbol{N}^{zz} = q^{zA} q^{zB} \boldsymbol{N}_{AB}$, which describes gravitational radiation through $\mathscr{I}$. In what follows, we shall specialise our discussion to the round-sphere conformal frame and set the super-translation mode to zero. We denote by γ the conformal factor appertaining to the round-sphere metric, so that $q_{AB}\mathrm{d}x^A \mathrm{d}x^B = \gamma^2 \mathrm{d}z\mathrm{d}\bar{z}$. Then, the physical celestial stress-tensor can be stated as

$$\boldsymbol{T}_{zz}^{\mathrm{CCFT}} = -\frac{1}{32\pi G_{\mathrm{N}}} \int_{\mathscr{C}} \frac{\mathrm{d}\bar{z}}{2\pi \mathrm{i}} \int_{\mathbf{R}} \mathrm{d}u \; u\gamma^2 \left(\mathrm{D}_z^3 - 2\mathcal{T}_{zz}\mathrm{D}_z - (\mathrm{D}_z \mathcal{T}_{zz})\right) \boldsymbol{N}^{zz} \,. \tag{3}$$

Note that, in the preceding equality, the contour integral $\int_{\mathscr{C}} \frac{\mathrm{d}\bar{z}}{2\pi \mathrm{i}}$ is utilised in order to extract the holomorphic celestial current density.

Now, recall that the holographic dictionary which we carefully explained in the preceding subsection exists within the context of the uplifting of the $\mathrm{AdS}_3/\mathrm{CFT}_2$ correspondence to flat-space holography, in the sense of Ball *et al.* [3]. Accordingly, following Nguyen [2], we shall argue that there is nothing in the geometry underlying this uplift which compels us to identify the physical celestial stress-tensor $\boldsymbol{T}_{zz}^{\mathrm{CCFT}}$ with the boundary-spacetime Virasoro stress-tensor $\boldsymbol{t}_{\mathrm{H}_3^+}$ originating from the string theory construction on Euclidean AdS_3 space. To see why, let $c_{\mathrm{H}_3^+}$ be the central charge of the boundary-spacetime Virasoro algebra of the underlying string-theory construction, and let $|\Psi\rangle$ be a boundary state whose associated semi-classical geometry belongs to the family of locally super-rotated vacua. Then, following Balasubramanian and Kraus [8] and Nguyen [2], it can be shown that

$$\frac{\langle \Psi | \boldsymbol{t}_{\mathrm{H}_3^+} | \Psi \rangle}{\langle \Psi | \Psi \rangle} \sim \frac{c_{\mathrm{H}_3^+}}{12} \mathcal{T}_{zz}[q_{AB}, \psi] \,, \tag{4}$$

which is non-zero for the non-trivial Schwarzian profiles, in the family of states to which $|\Psi\rangle$ belongs. In Eq. (4), we should evaluate $\mathcal{T}_{zz}$ on the geometric data $\{q_{AB}, \psi\}$, associated with the locally super-rotated vacuum corresponding to $|\Psi\rangle$, which we introduced above.

Therefore, as emphasised by Nguyen [2], the preceding result may be physically interpreted as saying that, under the uplifting of the AdS_3/CFT_2 correspondence to the flat-space holography, the classical expectation value of the boundary stress tensor $\boldsymbol{t}_{H_3^+}$, with respect to boundary states whose semiclassical geometry can be realised as a locally super-rotated vacuum, is mapped into $\mathcal{T}_{zz}$ rather than the physical celestial stress-tensor $\boldsymbol{T}_{zz}^{\text{CCFT}}$. Furthermore, the shear of the super-rotated vacuum is $C_{zz}^{\text{vac.}} = u\mathcal{T}_{zz}$, so that the covariant news becomes $\boldsymbol{N}_{zz} = \partial_u C_{zz}^{\text{vac.}} - \mathcal{T}_{zz} = 0$, since $\partial_u \mathcal{T}_{zz} = 0$. Hence $\mathcal{T}_{zz}$ may be non-zero whereas the vacuum remains non-radiative. (Remember that we specialised our discussion to the round-sphere conformal frame, and that we assumed that the super-translation mode is zero.)

Therefore, we have no reason compelling us to identify the holographic uplifting of the stress-tensor underlying the boundary-spacetime Virasoro algebra, originating from the string-theory construction, with the physical celestial stress-energy tensor. Consequently, no contradiction follows merely from the difference between their central charges.

III. THE QUESTION OF UNITARITY

I am grateful to the referee for encouraging the author to complement the manuscript with the discussion clarifying the non-unitarity of the candidate model for the perturbative celestial CFT based on the H_3^+-WZNW model by reviewing some notions appertaining to the WZNW theory. I am equally grateful to the reviewer for noticing that my reply to the first referee report indirectly indicated a connection between gauge theory and string theory on Euclidean AdS_3.

Remark. Interestingly, if I may briefly extend my discussion beyond the refereeing process, I would like to share with the reviewer and the editor that, in the past couple of months, I have been able to show that it is possible to construct a model for celestial CFT based on a generalisation of the $SL(N, \mathbf{C})/SU(N)$ WZNW model which contains a subset of vertex operators that, under an appropriate limit, generates the tree-level N^k-MHV celestial leaf amplitudes for gluons in $\mathcal{N} = 4$ SYM for all MHV degrees k. While I am still typesetting the results, the referee helped me to understand that this construction might be of interest beyond the celestial holography literature. At present, I have still not been able to find a string-theory completion or interpretation of my generalised version of the $SL(N, \mathbf{C})/SU(N)$ WZNW model in question, but I am convinced that its investigation shall be very illuminating.

I have accordingly followed the referee's suggestion and added the following discussion as subsection 1.2, "The Question of Unitarity."

Now, let me explain how the non-unitarity expected from the celestial CFT is compatible with the proposed holographic dictionary based on the H_3^+-WZNW model.

To make my argument unambiguous, I believe it should be helpful to begin by distinguishing some closely related but nonetheless distinct notions: the unitarity of the worldsheet-matter CFT appertaining to the H_3^+-WZNW model; the absence of reflection positivity underlying the celestial CFT, in the sense expected from conventional radial quantisation; and the unitarity of the action of $\mathrm{SL}(2,\mathbf{C})$ on the zero-mode Hilbert space $L^2(\mathrm{H}_3^+)$ of the H_3^+-WZNW model in the group-theoretic sense.

First, consider the following statement concerning the target space of the H_3^+-WZNW model:

$$\mathrm{H}_3^+ \cong \frac{\mathrm{SO}^+(1,3)}{\mathrm{SO}(3)} \cong \frac{\mathrm{SL}(2,\mathbf{C})}{\mathrm{SU}(2)}\,. \tag{5}$$

Geometrically, the meaning of the above assertion is that the target space underlying the H_3^+-WZNW model is a homogeneous space. Nonetheless, we should emphasise that the H_3^+-WZNW model belongs to a *different* class of conformal field theories than that defined by the conventional G/H coset construction.

In fact, in the ordinary vector-gauged realisation of a G/H coset CFT, we begin with a WZNW model with target space G and a global symmetry $G_{\mathrm{L}} \times G_{\mathrm{R}}$. Subsequently, we gauge a diagonally embedded subgroup $H \subset G_{\mathrm{L}} \times G_{\mathrm{R}}$. Now, by applying this construction with $G = \mathrm{SO}^+(1,3)$ and $H = \mathrm{SO}(3)$, we arrive at a model that does not have an $\mathrm{SO}^+(1,3)$ symmetry, because the symmetry described by $\mathrm{SO}^+(1,3)_L \times \mathrm{SO}^+(1,3)_{\mathrm{R}}$ is broken by gauging the diagonally embedded $\mathrm{SO}(3)$ subgroup. In other words, *the full original Lorentz symmetry does not survive as a global symmetry of the gauged model*.

On the other hand, the model defined by performing the analytic continuation of the $\mathrm{SU}(2)$ WZNW model is a sigma-model possessing H_3^+ as target space, which *does* have an $\mathrm{SO}^+(1,3)$ symmetry. Therefore, even if the H_3^+-WZNW model is a sigma-model whose target space is a homogeneous space $\mathrm{SO}^+(1,3)/\mathrm{SO}(3)$ or $\mathrm{SL}(2,\mathbf{C})/\mathrm{SU}(2)$, it is not an $\mathrm{SO}^+(1,3)/\mathrm{SO}(3)$ or $\mathrm{SL}(2,\mathbf{C})/\mathrm{SU}(2)$ coset model in the sense of the conventional vector-gauged realisation of the coset construction, explained, for example, in Di Francesco *et al.* [9]. See, for example, subsection 5.2 of Murthy and Witten [10] or the introduction of Gawedzki [11].

The physical significance of the preceding observations is that the familiar unitarity properties underlying the coset CFTs, originating from compact WZNW models at positive integer levels, should not be inferred merely from the fact that the target manifold is a homogeneous space. In especial, the worldsheet-matter CFT appertaining to the H_3^+-WZNW model, on which our proposed holographic dictionary is based, is *non-unitary*. The reason is that the H_3^+-WZNW model can be understood as originating from the analytic continuation of the $\mathrm{SU}(2)$ WZNW model, and the procedure of analytic continuation multiplies the Wess-Zumino term by the imaginary unity, thereby making that contribution imaginary when the worldsheet has Lorentzian signature, a point emphasised in section 5 of Murthy and Witten [10]. The resulting worldsheet-matter CFT thereby lacks the reality properties required for conventional unitarity

Note that there exists no contradiction between the above remarks and the positivity of the zero-mode sector of the H_3^+-WZNW model. Recall that the Hilbert state-space

of the zero-mode sector is

$$\mathscr{H}_{\text{zero-mode}} = L^2(\mathrm{H}_3^+, \mathrm{d}h)\,, \tag{6}$$

and carries a unitary $\mathrm{SL}(2,\mathbf{C})$ representation. However, the contribution originating from the WZ term vanishes in the zero-mode truncation. Thus, positivity of the zero-mode inner product does not imply reflection positivity of the complete worldsheet-matter CFT.

IV. SPELLING AND TYPOGRAPHICAL ERRORS

I must thank the referee for noting the grammatical and typographical errors in the manuscript. I must confess that, while Fernando Pessoa wrote impeccable English verse, the present author shares Pessoa's mother tongue but not his literary talents, and I therefore thank the referee for their patience! Below are the errors I found using the program "LanguageTool" for LaTeX. Please note that the page numbers in the following errata refer to the original manuscript, not to the new version I am submitting.

1. Page 4: "the hyperboloid ... correspond to" → "the hyperboloid ... corresponds to".
2. Page 8: "lacks the reality properties required for conventional unitarity" → "lacks the reality properties required for conventional unitarity."
3. Page 11: "specially those by" → "especially those by".
4. Page 16: "Recalling that ..., the KZ equations for the currents ... take the form" → "Recalling that ..., we find that the KZ equations for the currents ... take the form".
5. Page 19: "throughly analysed" → "throughly analysed".
6. Page 26: "a similar structure as Eq. (67)" → "a similar structure to that of Eq. (67)".
7. Page 30: "the product of gamma functions ... are not merely proportionality constants. Rather, they are physical quantities that characterise the scattering processes." → "the product of gamma functions ... is not merely a proportionality constant. Rather, it is a physical quantity that characterises the scattering processes."
8. Page 31: "Using the Sugawara construction (reviewed in ...), the stress tensor ... satisfies the Ward identity" → "Using the Sugawara construction (reviewed in ...), we find that the stress tensor ... satisfies the Ward identity".
9. Page 32: "the KZ equations takes the form" → "the KZ equations take the form".
10. Page 34-35: "Recalling ... that [Eq. (141)], the partial derivative with respect to x yields" → "Recalling ... that [Eq. (141)], we take the partial derivative with respect to x to obtain."
11. Page 37: "a broader research program" → "a broader research programme."
12. Page 38: "orbifolds solutions" → "orbifold solutions".

13. Page 46: “This transformation allow us” → “This transformation allows us.”

14. Page 48: “we are lead to define” → “we are led to define”.

15. Page 51: “Using the identities from (see Giveon, Kutasov, and Seiberg [13]), namely” → “Using the identities from Giveon, Kutasov, and Seiberg [13], namely”.

16. Page 55: “The method of Feynman α-parameters instruct us” → “The method of Feynman α-parameters instructs us”.

17. Page 58: “its interesting” → “it’s interesting”.

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[1] L. Donnay and R. Ruzziconi, Journal of High Energy Physics **2021**, 1 (2021).
[2] K. Nguyen, PoS **CORFU2021**, 133 (2022), arXiv:2201.09640 [hep-th].
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[9] P. Di Francesco, P. Mathieu, D. Sénéchal, and P. Mathieu, *Conformal field theory*, Vol. 24 (Springer New York, 1997).
[10] S. Murthy and E. Witten, Communications in Mathematical Physics **407**, 154 (2026).
[11] K. Gawedzki, in *New symmetry principles in quantum field theory* (Springer, 1992) pp. 247–274.